Joe (Nazhou) LIU

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Education

Nanyang Technological University Singapore Aug 2021 – May 2025
Bachelor of Engineering in Electrical and Electronic Engineering *Singapore*

- **Grade**: **Distinction**
- **Specialization**: Machine Learning
- **NTU President Research Scholar**: Awarded to outstanding undergraduates to pursue research projects

Publications

[1] **N. Liu**[*], Y. Fei[*], X. Yu[*], S. Chen, L. Li, R. Thapa, M. Ciobanu, N. P. Singh, Q. Mao, R. Das. *How Do Agents Fail on AutoResearch: End-to-End Diagnostic Evaluation on 100 Real-World Frontier Research Tasks.* arXiv:2608.14905, 2026. **IAB Workshop @ NeurIPS 2026** (non-archival); under review, **ACL 2027**. ([*] equal contribution) **[arXiv]**

[2] C. Li, **N. Liu**, K. Yang. *Adaptive Group Policy Optimization: Towards Stable Training and Token-Efficient Reasoning.* arXiv:2503.15952, 2025. **(28 citations) [arXiv]**

[3] Y. Fei, L. Li, S. Chen, **N. Liu**, X. Yu, M. Ciobanu, N. P. Singh, Q. Mao, R. Das. *MatHarness: Towards Long-Horizon Open-ended AutoResearch for Computational Materials Science.* **AI4Mat Workshop @ NeurIPS 2026**.

[4] S. Chen, Y. Fei, **N. Liu**, X. Yu, L. Li, R. Thapa, M. Ciobanu, Q. Mao, R. Das. *Reconstruction: A Blind Benchmark for Recovering Research Ideas from Pre-Publication Bibliographies.* arXiv:2608.16645, 2026. Under review, **Nature Machine Intelligence**. **[arXiv]**

[5] "A Text-Assisted Reading Method, Device, Equipment, Medium and Program Product", Tong Li, Chen Li, Tong Jia, Yuting Zhong, **Nazhou Liu**, Siqi Mai, Hongxin Liu, Zhengda Bian, Yongbin Li, Ziyuan Cui, Yuanheng Zhao, Patent Application CN119293168A, Jan. 10, 2025, Beijing Luchen Technology Co., Ltd. **[patent]**

Professional Experience

Prentis AI Apr 2026 – Now
Member of Technical Staff *Shanghai*

AutoResearchEval & ARFT — Diagnosing Failures of AutoResearch Agents

- Co-first-authored **AutoResearchEval**, an open-source collection of **800** agent trajectories with artifact-aware, process-level failure annotations, released with the underlying **100**-task, **7**-domain full-lifecycle task suite built from published frontier science, including a fully open-ended subset scored on process rather than outcome **[arXiv]**
- Built **ARFT**, the first systematic failure taxonomy for autonomous research agents, organizing **45** empirically grounded failure patterns along two cross-cutting axes—the lifecycle stage at which a failure manifests and its underlying root cause—so that superficially different failures sharing a mechanism group together
- Developed the human-calibrated, artifact-aware **agent-as-a-judge** annotator behind ARFT categorization, reading each trajectory in full—run logs, generated data, code and reports—rather than its final answer alone

MatHarness — Long-Horizon AI Researcher Agent

- Built **MatHarness**, an end-to-end **AI researcher agent** spanning ideation to manuscript writing, coupling **execution robustness** (pre-flight verification, expert-distilled skills, just-in-time auditing), **epistemic adaptability** (persistent research artifacts and an evolving scientific mental model), and a **heterogeneous multi-agent round table** under targeted human oversight
- Validated end-to-end on a real **AI-for-science** problem in computational chemistry: over a **multi-week** autonomous campaign the system revised its own research plan **6** times, recovered from failed computations, and had its auditor overturn one of its own over-attributed claims before publication—domain collaborators estimated the same campaign would occupy an experienced PhD researcher **1–2 months**, against the system's **∼2 weeks**

HPC-AI Tech Feb 2025 – Apr 2026
AI Algorithm Engineer *Singapore, Shanghai*

HPC-AI Fine-tuning SDK

- Co-developed "Post-Training as a Service" platform by replicating **Tinker** framework. The platform supports scalable SFT and RL for text-based LLMs and Multimodal LLMs using memory-efficient **LoRA fine-tuning**

- Led the development of core training primitives (**forward/backward**, **optimizer steps**), integrating optimization techniques such as Sequence Unpacking, Gradient Checkpointing, and Mini-batching; significantly increased GPU compute throughput by maximally **35%** during large-batch training
- Architected co-located RL system that unifies **FSDP-based training** and **vLLM-powered sampling** on shared GPU resources; eliminated up to **50%** of training GPU idle time during sampling phase and drastically reduced inter-node communication latency

Colossal-R1

- Experienced in cutting-edge research on **reinforcement learning** for reasoning LLM, applying algorithm refinement techniques such as curriculum learning, multi-stage learning, resulting in significant improvement on benchmarks
- Proposed new RL algorithm **Adaptive Group Policy Optimization(AGPO)** to replace GRPO, ensuring more stable policy entropy during training and achieving better performance with fewer CoT token usage during inference time [**arXiv**]
- Developed data collection and verification pipeline to generate high quality math and code data for RL training, ensuring balanced difficulty and sub-categories coverage across dataset
- Conducted large-scale RL training across multi-GPU clusters using **verl** framework, boosting AIME24 accuracy by **10%** and LiveCodeBench scores by **11.4%** over baseline models [**Related News**]

HPC-AI Tech May 2024 – Nov 2024

AI Algorithm Engineer Intern *Singapore*

Colossal-Reader

- Contributed to open source project Colossal-AI (with more than **41k stars** on GitHub), focusing on **RAG** system design and document **QA chatbot** implementation
- Refactored parsing, chunking and generation modules of RAG system, increasing accuracy by **12%** on internal end-to-end evaluation dataset; multi-level segmentation and query routing filed as a patent application [**patent**]
- Upgraded original RAG system to **Multi-modal RAG** system through supporting **VLM** generation, increasing response accuracy on PDFs by **28%**

Central Provident Fund Board May 2023 – Aug 2023

AI Engineer Intern *Singapore*

CPF Bot

- Implemented **RAG** system and **LLM agents** a year before it was trending, aiming for assisting customer service staff to answer CPF-related questions
- Designed and improved LLM-powered **ReAct** agent, reducing hallucination rate by **15%** through prompt iteration and fact-checking tool calling

Skills

LLM Post-Training: SFT, **RLHF**, **DPO**, **RLVR** and **OPD**, in both **LoRA** and **full-parameter** distributed settings. Grounded in the theory—objective derivations, policy-gradient estimators, importance sampling, and entropy control. Multiple experiences improving the algorithms themselves, including alrotirhms such as **GRPO**, and **OPD**.

Agents & Harness Building: Multiple agents built, from **ReAct** Agents to **Autoresearch** Agents to **RSI** Agents. Familiar with agent **failure patterns** and the design philosophy behind coding agents such as **Claude Code**. Build **long-horizon** agents from **first principles**: the harness as **context management**—controlling what the agent sees and knows.

LLM Infrastructure: Comfortable at LLM Infra—asynchronous RL frameworks (**verl**, slime), high-throughput inference (**vLLM**, SGLang), and distributed training (**PyTorch FSDP**/DDP, DeepSpeed). Experienced building **robust training pipelines** and integrating optimization techniques—sequence unpacking, gradient checkpointing, mini-batching, asynchronous training and sampling—for better stability and efficiency.

LLM Architecture & Pretraining: Theoretical understanding of modern LLM design—decoder-only architectures , attention variants (**GQA, MLA, AttnRes**), **MoE** routing, **RoPE** and **scaling laws**.

Languages: Fluent in English and Chinese (Mandarin)

Service

Reviewer, Interpreting Agent Behavior (IAB) Workshop @ **NeurIPS 2026**
Main Committee, Institution of Engineering and Technology (IET), NTU, 2023–2025